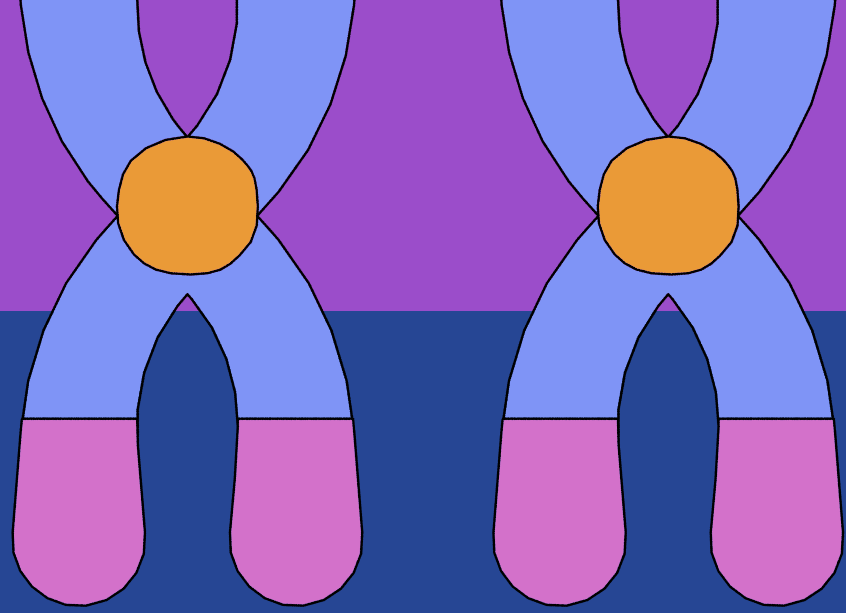
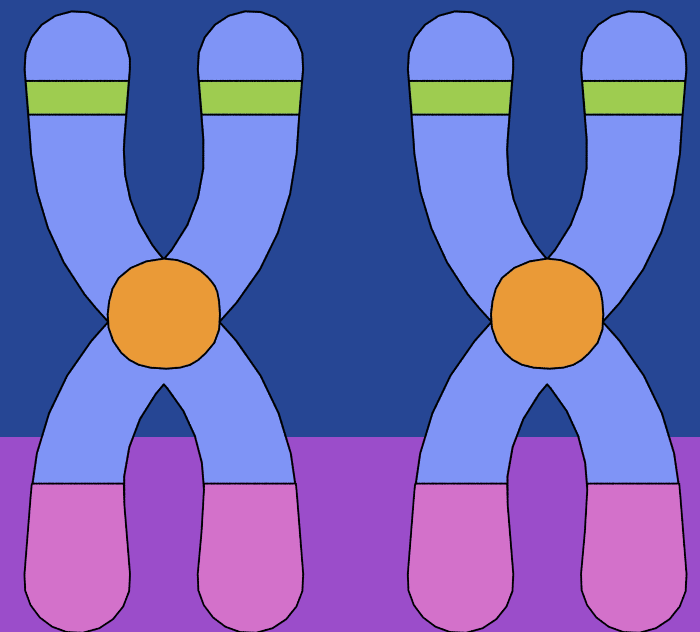




Cancer Type Prediction Using Gene Expression Data

prepared by
Alaa Khaled Shukair
ID:221002049

CBIO313 – Data Mining and Machine
Learning Spring 2026





Project Objective

- Predict cancer type using gene expression data
- Apply preprocessing and feature engineering
- Compare multiple machine learning algorithms
- Deploy the final model using Streamlit

Project question

Can machine learning predict the cancer type of a sample using gene expression data?

Dataset Overview

	Id	gene_1	gene_3	gene_5	gene_7	gene_8	gene_9	gene_10	gene_11	gene_13	...	gene_20634	gene_20635	gene_20636	gene_20637	gene_20638	gene_20639	gene_20640	gene_20641
0	sample_664	0.160738	-0.327348	-0.144638	0.196493	-1.105093	0.309926	-0.177461	-1.124182	-0.459826	...	-1.611378	-1.108411	-0.670719	-1.739299	0.476467	1.136071	-0.576601	-1.275518
1	sample_215	-0.771173	0.885819	-0.234209	0.273139	0.132208	-0.249541	0.005817	-0.631647	NaN	...	0.247812	0.144035	0.148776	-1.373208	0.099245	0.391993	0.573363	0.322198
2	sample_343	-0.169258	1.908618	0.165008	-0.562826	0.199720	0.128036	2.348450	2.425346	-0.933545	...	1.133065	0.965014	1.873753	-0.005167	-0.223091	0.782868	-0.562787	-0.471593
3	sample_707	-0.947912	0.111177	-0.153179	0.837412	0.185467	-0.066223	-0.267734	0.674365	-0.076086	...	0.022339	0.326506	-0.333964	0.228595	-0.245309	0.478564	0.273364	1.756369
4	sample_621	-0.335741	0.515251	0.325440	-0.842387	-0.500415	0.484240	-0.438587	-0.874562	NaN	...	-1.516812	-1.430622	-0.664933	-0.753410	NaN	0.375521	-0.536705	-0.523850

5 rows × 14574 columns

- Dataset contains gene expression values
- 150 samples and 14,574 features
- Multi-class cancer classification problem
- Real-world biomedical dataset

Missing Values & Data Cleaning

```
New dataset shape: (150, 14574)
```

```
Remaining missing values in Class:  
0
```

```
Columns with missing values: 14573
```

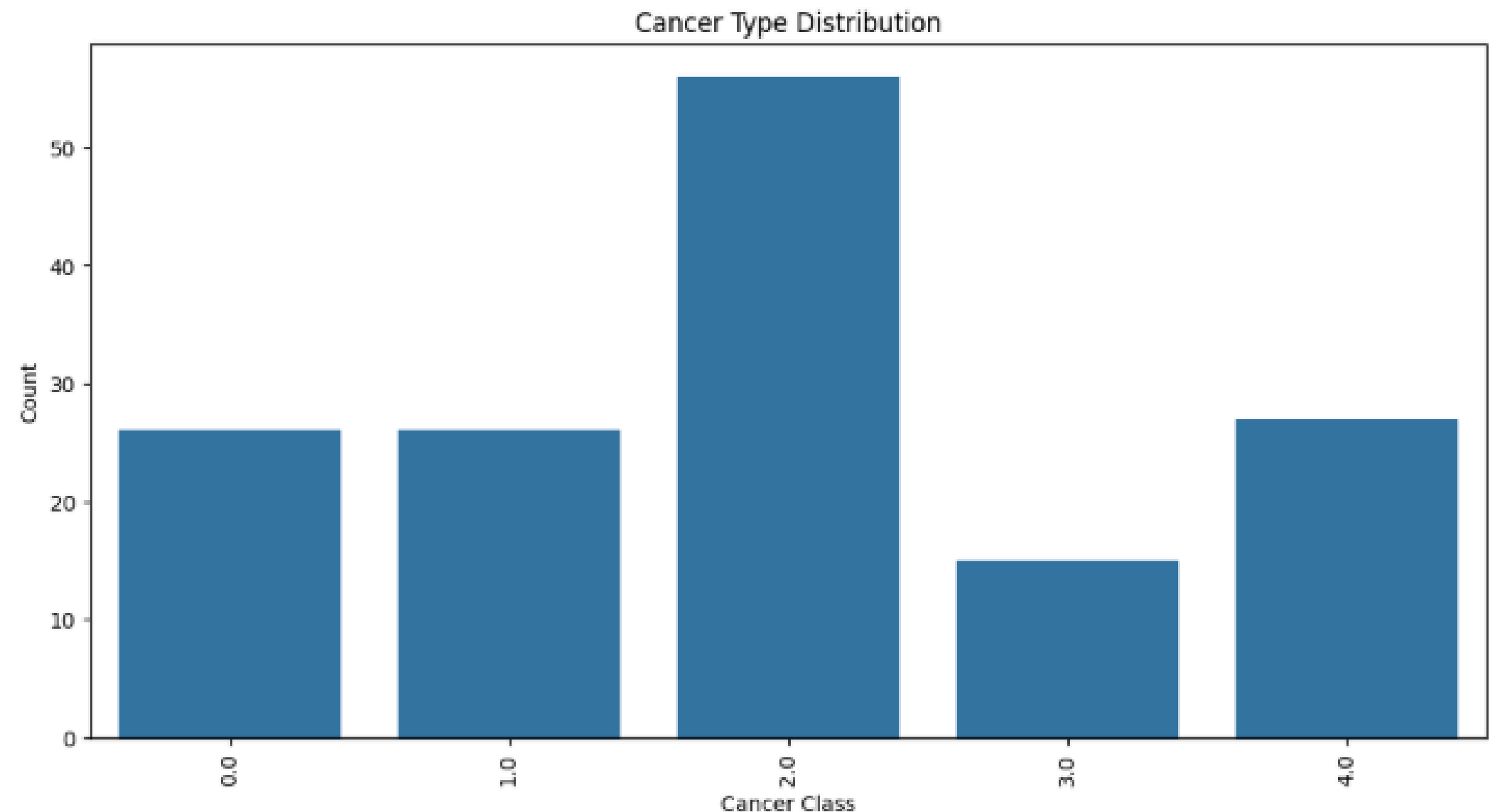
```
Top missing values:
```

Class	250
gene_7972	59
gene_20314	54
gene_6045	54
gene_579	54
gene_3144	53
gene_8152	53
gene_19033	53
gene_7234	52
gene_6489	52
dtype:	int64

- Missing values detected in many genes
- Missing values handled during preprocessing
- Dataset cleaned successfully
- Final dataset ready for analysis

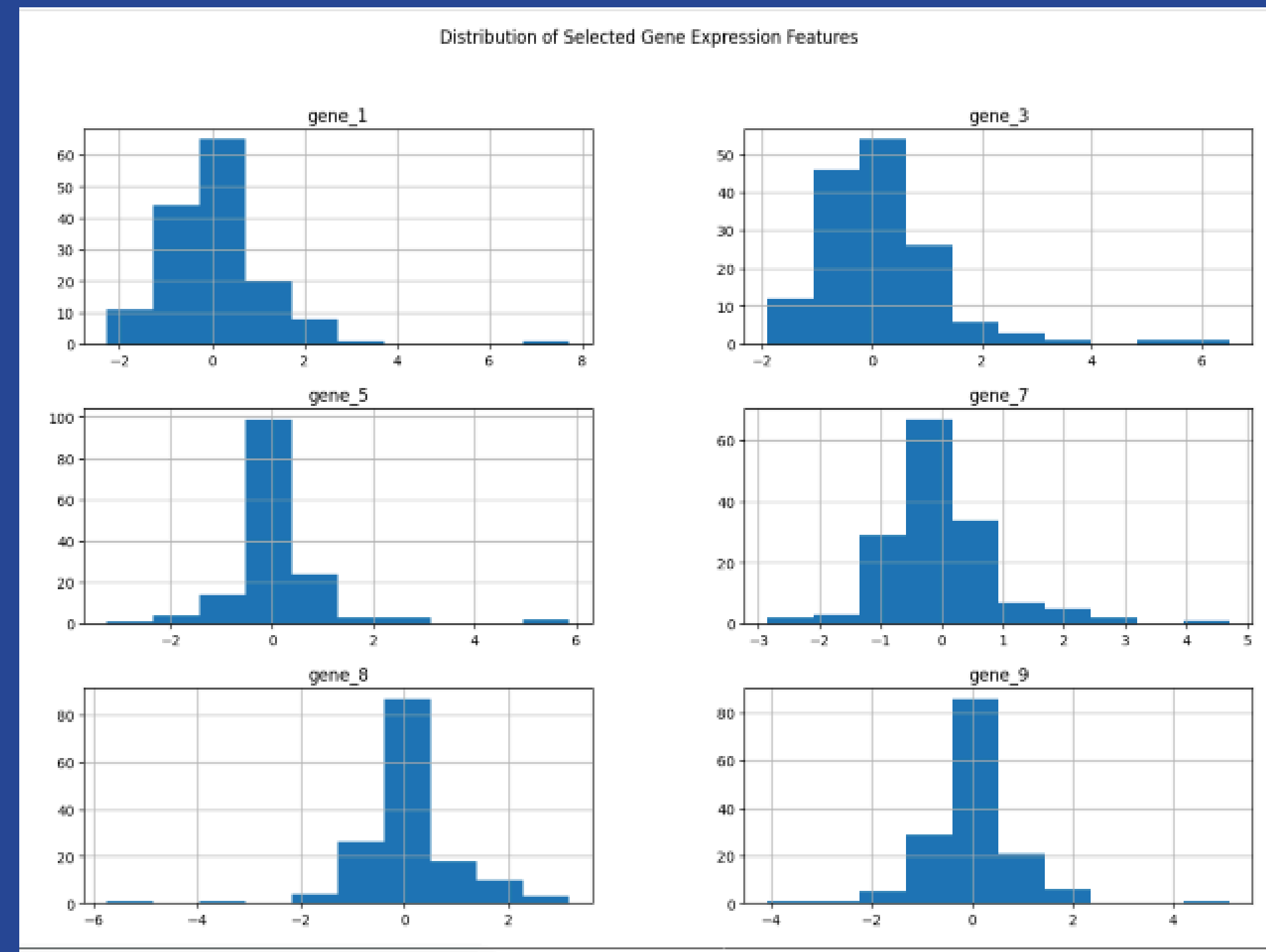
Exploratory Data Analysis

- Class distribution analyzed
- Some classes contain more samples than others
- Dataset shows moderate imbalance



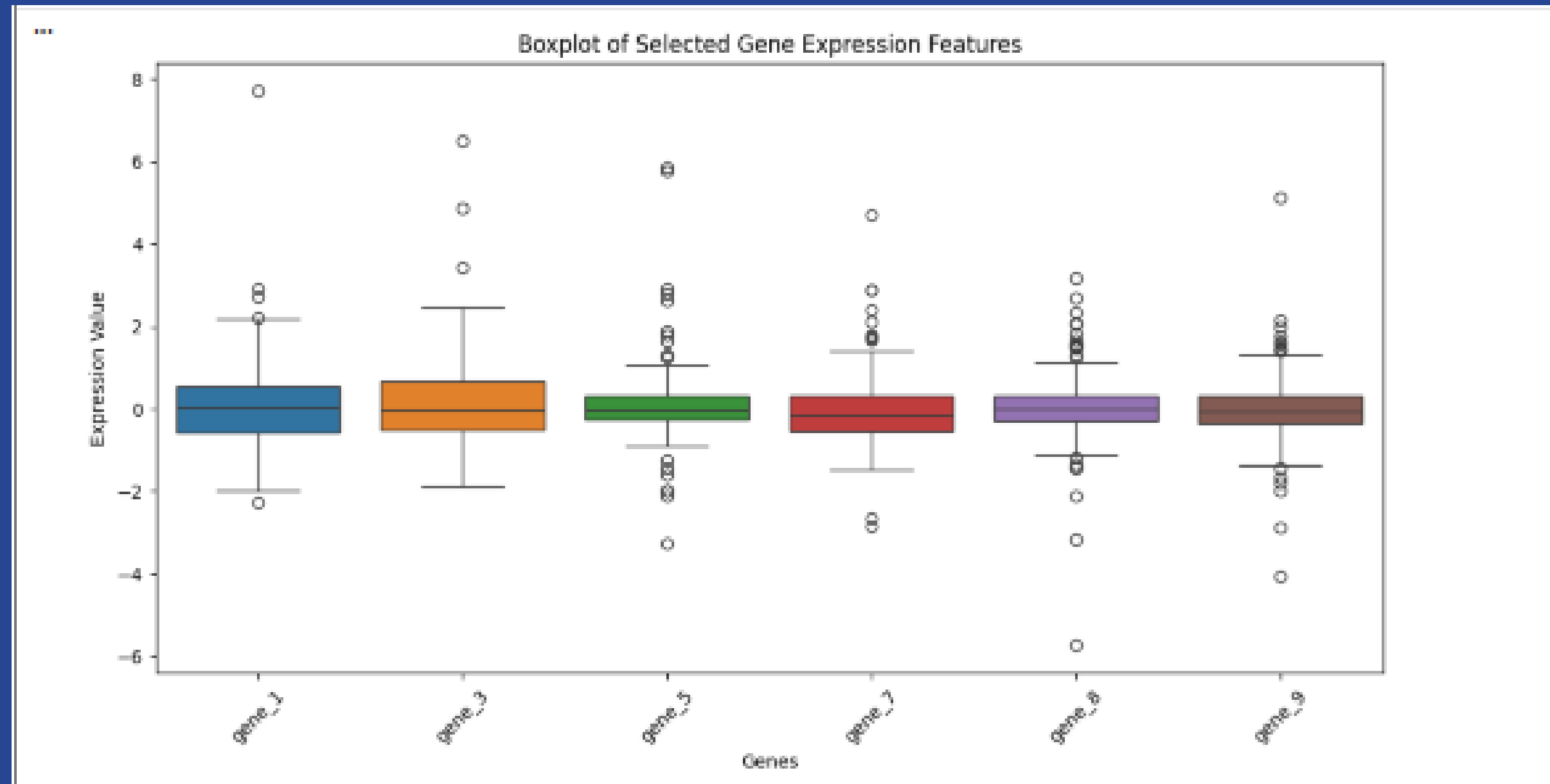
Feature Distribution

- Distribution of selected gene expression features
- Most genes centered around zero
- Presence of outliers and variability



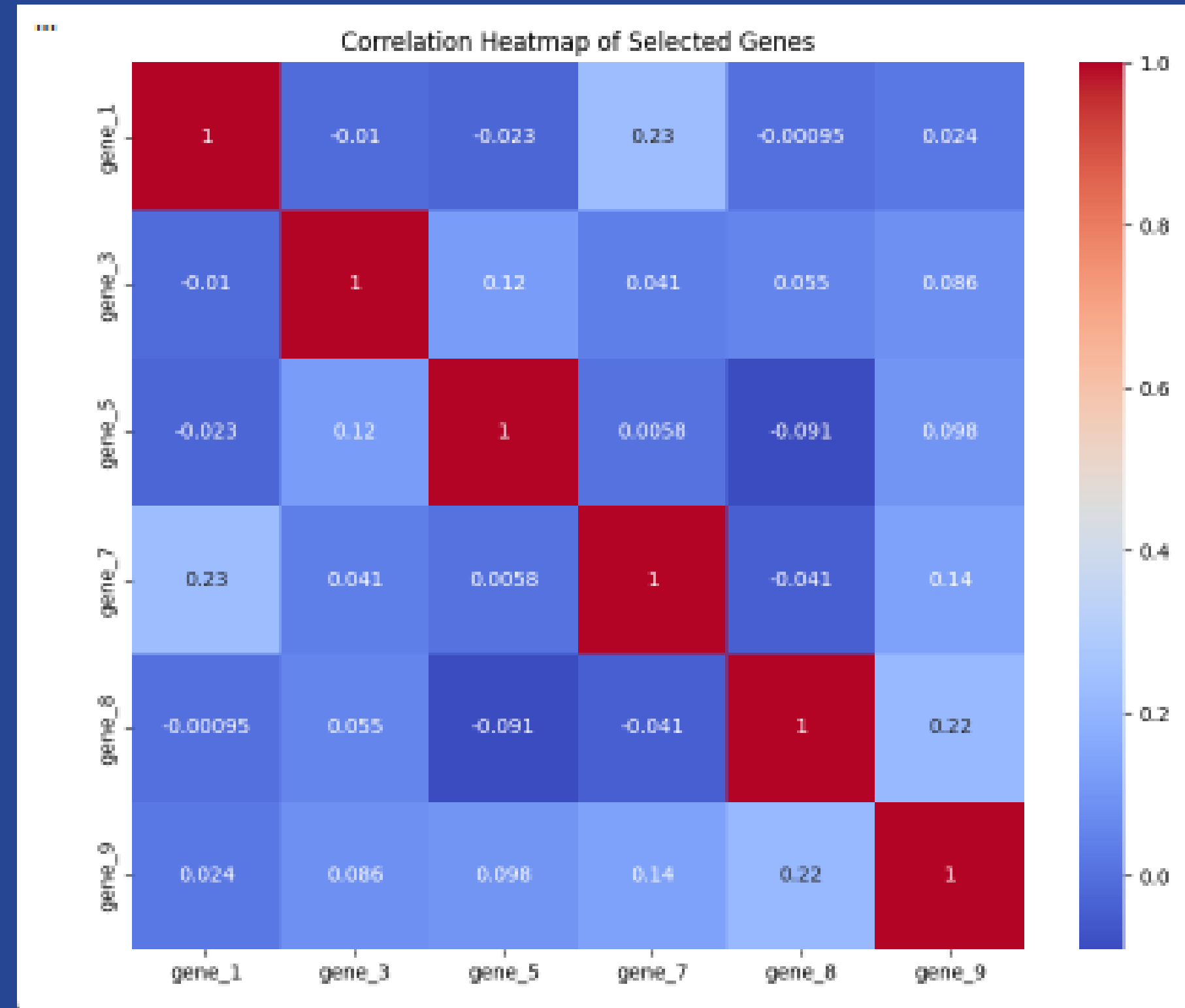
Boxplot Analysis

- Outliers detected across several genes
- Gene expression values show variability
- Important for understanding data spread



Correlation Analysis

- Relationship between gene features visualized
- Different cancer classes overlap partially
- Indicates classification complexity



Feature Engineering

- Created new statistical features
- Mean gene expression
- Standard deviation of gene expression
- Improved feature representation

Updated dataset shape after feature engineering: (150, 14574)

	mean_gene_expression	std_gene_expression
0	0.039442	0.588071
1	0.013771	0.867731
2	0.010345	0.893224
3	0.023744	1.009574
4	-0.095743	0.628603



Feature Selection

- SelectKBest with ANOVA F-test used
- Reduced features from 14,574 to 100
- Helps reduce dimensionality
- Improves model efficiency

```
Original shape: (150, 14574)  
Selected shape: (150, 100)
```

Train/Test Split

- Dataset split into training and testing sets
- StandardScaler applied
- Prevented data leakage
- Improved model performance

```
Training features shape: (120, 100)
Testing features shape: (30, 100)
Training target shape: (120,)
Testing target shape: (30,)
```

```
|| =====
```

```
Scaling completed successfully!
```

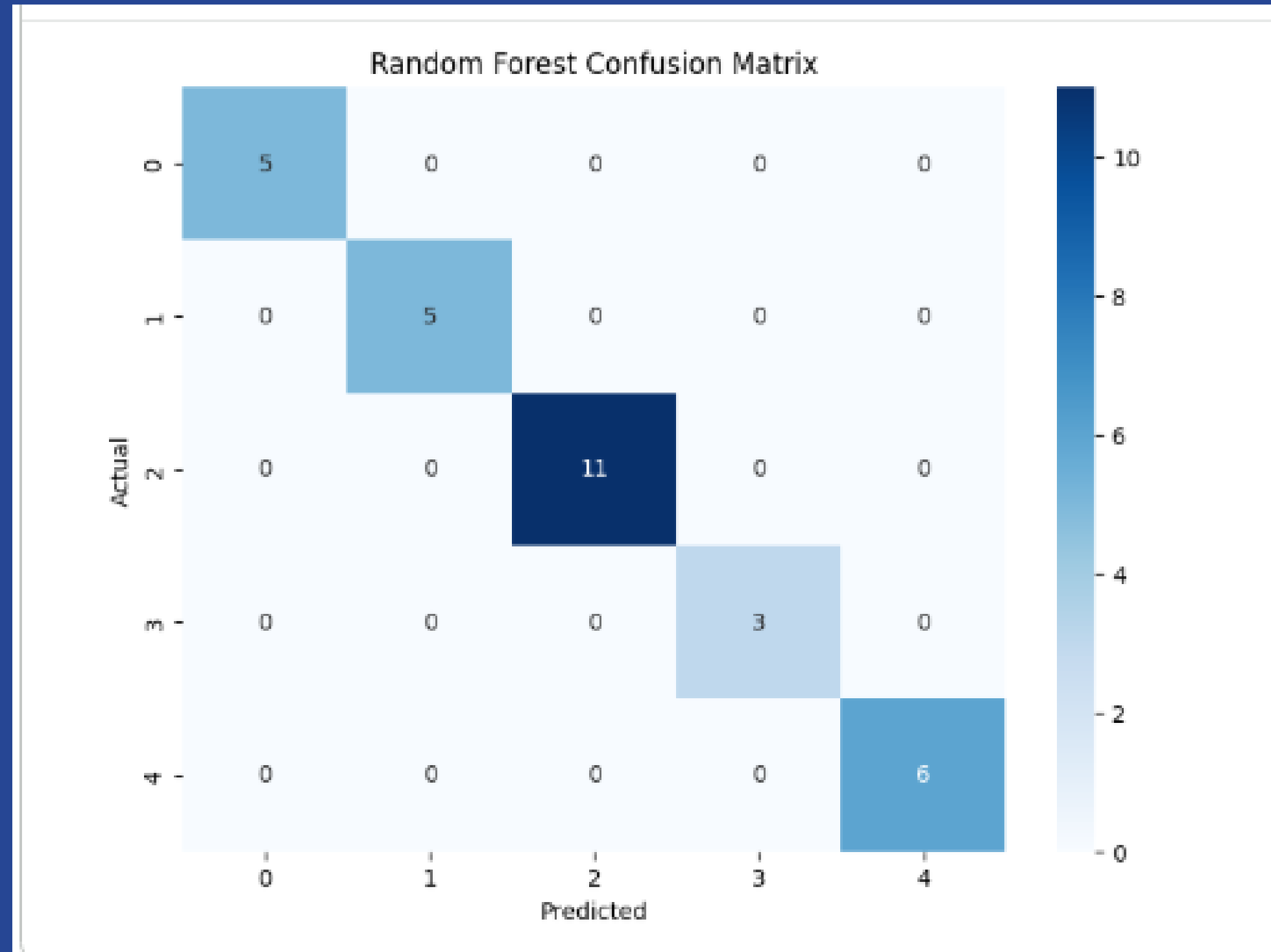
Machine Learning Models

- Logistic Regression
- Random Forest
- Support Vector Machine
- Models compared using multiple metrics

results						 
	Model	Accuracy	Precision	Recall	F1 Score	
0	Logistic Regression	1.0	1.0	1.0	1.0	
1	Random Forest	1.0	1.0	1.0	1.0	
2	Support Vector Machine	1.0	1.0	1.0	1.0	

Best Model Evaluation

- Random Forest achieved best performance
- High classification accuracy
- Excellent class prediction capability



Classification Report

- Precision, Recall, and F1-score evaluated
- Strong performance across all classes
- Balanced model behavior

	precision	recall	f1-score	support
0	1.00	1.00	1.00	5
1	1.00	1.00	1.00	5
2	1.00	1.00	1.00	11
3	1.00	1.00	1.00	3
4	1.00	1.00	1.00	6
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Hyperparameter Tuning

Best Parameters:

```
{'max_depth': None, 'min_samples_split': 2, 'n_estimators': 100}
```

Best Cross Validation Score:

```
0.9916666666666667
```

- GridSearchCV used for optimization
- Tuned Random Forest parameters
- Improved cross-validation performance

Streamlit Deployment

- The final model was deployed using Streamlit.
- The user uploads an Excel/CSV sheet containing gene expression values.
- The app preprocesses the uploaded gene data using the saved scaler and selected features.
- The trained model predicts the cancer class automatically.
- The prediction result is displayed directly on the web app.

Cancer Type Prediction Using Gene Expression Data

Upload a CSV file containing gene expression values to predict the cancer class.

The uploaded file should contain the same gene columns used during model training.

Upload gene expression CSV file

 class3_..._clean.csv 426.0KB +

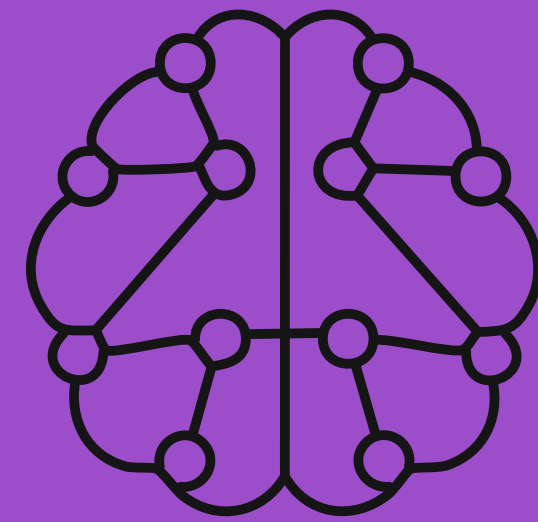
Uploaded Data Preview

	gene_1	gene_3	gene_5	gene_7	gene_8	gene_9	gene_10	gene_11	gene_13	gene_14	gene_15
0	-0.7274	0.2186	0.0293	-0.5819	0.0553	-0.371	-0.4774	-0.4058	-0.1266	-0.2065	1.2

Prediction Result

Predicted Cancer Class: 3.0

Conclusion



- Successfully predicted cancer classes
- Feature selection improved efficiency
- Random Forest achieved best results
- Streamlit deployment completed successfully